Chapter 11

Evidence-Based Practice: Use in Answering Queries and Developing Systematic Reviews

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Learning Objectives

Objective 11.1 Define evidence-based practice and explain its framework.
Objective 11.2 Recognize and employ the steps for the modified systematic approach when answering drug information-related questions.
Objective 11.3 Describe the process of preparing systematic reviews using evidence-based practice.

OBJECTIVE 11.1. DEFINE EVIDENCE-BASED PRACTICE AND EXPLAIN ITS FRAMEWORK

Evidence-based practice (EBP) is the integration of best research evidence, clinical expertise, and patient values. EBP is more than theory and skills; it is a way of practicing pharmacy/medicine and therefore should be integrated into the culture of an institution. For these skills and culture to solidify, it is essential that pharmacists learn the concepts of EBP during their education and carry on this practice throughout their professional careers.

The goal of EBP is to achieve optimal patient outcomes by integrating the best research evidence available, along with clinical expertise and patient values/preferences when making patient-care decisions.[1,2] With that in mind, it is important to acknowledge that using a systematic approach to finding the best available research evidence is essential. Overall, five distinct steps create the framework for EBP:

1. Ask: Organizing the clinical question in a structured format is important when developing a literature search. Using a format known as PICO(T) promotes retrieval of articles more closely linked to the clinical question (Table 11.1).
2. Search: Using research databases effectively is dependent on the utilization of concepts and keywords in combination with each other to filter out articles that do not fit the clinical question.
3. Critical Appraisal: Evaluating research articles related to the clinical question of interest can be efficiently completed by focusing on three key questions, which are presented in depth later in this chapter:
   a. *Are the results valid?* The emphasis here is on the design of the study and any limitations that might impact the results.
   b. *Are the results significant?* If the results are valid, it is essential to evaluate whether the results of the study are statistically significant.
   c. *Will these results help determine care for the patient*? Finally, decide how the study's findings can help treat the patient. This may be more related to the clinical relevance of the findings.
4. Implement: Creating a clinical "action plan" based on evidence from the literature is often an underappreciated component of the EBP process. However, achieving and maintaining success without a formal plan that relies on establishing accountability, support of key clinical stakeholders, and a standardized process is very difficult. This is also the ideal step in the process to fully integrate patient values and preferences through shared decision-making.
5. Evaluate: Following implementation, it is important to monitor and evaluate patient and quality outcomes.

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TABLE 11.1 PICO(T)—Formulating an Answerable Question

Description	Example
P: Patient/population or problem The patient is a member of a population with a specific health problem. However, factors such as age, sex, and ethnicity should also be considered.	Hospitalized adult patient with recurrent methicillin-resistant *Staphylococcus aureus* bacteremia
I: Intervention The therapy being considered or evaluated.	Linezolid
C: Comparison The "standard care" or comparison therapy.	Vancomycin
O: Outcome The desired outcome should be measurable.	Effective infection clearance
T: Time (if important)	

Many clinicians and healthcare leaders agree that a sixth step, dissemination, also plays an integral role in EBP. This is because dissemination of new knowledge through peer-reviewed publication not only fosters support of the EBP process but also promotes improvements in clinical practice as the literature base for decision-making evolves.

It is also important to think about the use of the EBP in two distinct ways[3]:

- Evidence-based individual decision-making—where the focus is individual clinicians using research evidence when facilitating decision-making to treat individual patients (i.e., a septic patient with urinary tract source and documented penicillin anaphylaxis).
- Evidence-based guidelines—where the focus is using the best research evidence to make recommendations about how to treat a group of patients defined by a specific clinical criterion (i.e., urinary tract infection), rather than an individual patient.

Although the use of best research evidence is an important factor in decision-making for both of these practices, they are not mutually exclusive. Both methodologies play a vital role in improving clinical practice and patient outcomes.

Other Methods Used in EBP

The modified systematic approach explained in depth in this chapter is one of many methods used to promote the integration of research evidence into practice. Another familiar EBP methodology is Critically Appraised Topics (CATs). The Centre for Evidence-Based Medicine at University of Oxford in the United Kingdom has developed standardized templates for CATs in multiple languages that can be used to critically appraise systematic reviews, randomized controlled trials (RCTs), and observational studies on diagnosis and prognosis[4,5] The Centre for Evidence-Based Medicine at University of Oxford in the United Kingdom has also created a downloadable critical appraisal tool, called CATmaker,[5] which supports the step-by-step use of EBP by prompting users to:

- Ask a focused and answerable question
- Search for the best available evidence
- Critically appraise the evidence for validity and clinical relevance
- Apply the results to clinical practice
- Evaluate performance

OBJECTIVE 11.2. RECOGNIZE AND EMPLOY THE STEPS FOR THE MODIFIED SYSTEMATIC APPROACH WHEN ANSWERING DRUG INFORMATION-RELATED QUESTIONS

Use of EBP comes into play in a variety of different ways for clinicians. Healthcare professionals answer a wide variety of questions from their patients, as well as other colleagues. When researching an answer, pharmacists are often sought out for their experience and expertise about specific questions relating to medication use and safety. Whether one is a hospital

TABLE 11.2 Seven Principles of the Modified Systematic Approach[7,9]

1. Secure the demographics of the requester and determine when the information is needed
2. Retrieve the necessary background information from the requester about the question
3. Determine the true informational need and categorize the ultimate question to be answered
4. Develop an appropriate searching strategy and conduct a thorough search
5. Evaluate, analyze, and synthesize the gathered information
6. Formulate and communicate the answer
7. Conduct follow-up, if necessary, and provide documentation

pharmacist stationed in the medical intensive care unit, a retail pharmacist working at an independent or chain drugstore, or a long-term care pharmacist making rounds through a specialized care home, the pharmacist is seen as a valuable source of drug information. For example, a patient receiving warfarin for the first time may experience bruising, and a nurse may want to know the specific monitoring parameters (e.g., blood work, INR) for the patient's specific diagnosis. A medical resident may initially prescribe clindamycin for a mild skin infection in a patient allergic to erythromycin and wants to know if it is safe to do so. A psychiatrist may be treating a patient for depression with selective serotonin reuptake inhibitors and may inquire about starting or stopping other medications to minimize the potential for drug-drug interactions. Whatever the case, the pharmacist should know the best way to search the literature using EBP to answer the question efficiently and effectively.

Pharmacy schools that have a drug information course or teach the concept of providing drug information throughout the curriculum usually employ the use of the modified systematic approach to answering specific drug information questions. This method has been developed by Watanabe[6] in 1975 but has been published and modified to its current format by other noted instructors within drug information practice.[7,8] Basically, it centers on seven principles and employs the basic tenants of using EBP to validate and verify the information researched (Table 11.2).

Secure the Demographics of the Requester

This seems like a fairly simple thing to do, but actually, it is probably one of the most important aspects of the modified systematic approach. Securing the demographics of the requester will set the stage for how to approach the question, what format to use in answering the question, and when the information is needed by the requester.[8,10]

For a question over the phone, the name of the person calling and the number where they can be reached are the first two items that should be collected. Without knowing who is being addressed and how to ultimately communicate the final response to the person calling, the rest of the principles of the modified systematic approach will prove futile. Therefore, it is extremely important to take down the correct information regarding the person's name, their title (e.g., nurse, physician, pharmacist, patient), and contact information (i.e., fax/phone number). A person's title is very important to how the question will be answered because the tone and wording of the answer will vary accordingly. Additionally, some people may wish to receive the information via email and then respond with further questions if the answer is not clear or more information is desired. Email addresses can be cumbersome, especially if they are associated with a private email account. Be sure to repeat email addresses and fax/phone numbers back to the requester to ensure that the right contact information has been collected.

This would also be the best time to ask when the information is needed. Common responses from requesters when this question is posed are "Whenever you can get to it," "As soon as possible," or sometimes, a specific date and time is given. It is important to narrow down the time frame with the first two examples. If the requester says, "Whenever you can get to it," then time may not be of the utmost importance. As long as it is not a scenario where the patient is currently in the hospital, in the clinic, in the counseling area, or waiting for their prescription, one might be able to offer a time in the near future for the answer to be provided (e.g., 24 h, next week). This would allow additional or necessary time for research, especially if the question is a particularly in-depth one (e.g., various secondary databases and primary literature will need to be searched). If the requester indicates that the answer is needed as soon as possible, education of the person making the request may be necessary. Most drug information centers do not provide drug information within 30 min or less because more time is often essential for education, searching, formatting, and editing. In addition, most healthcare professionals

who use drug information centers understand that a large amount of work and effort might be required to find the answer to an unusual or peculiar question. Finally, most pharmacies (e.g., hospital, retail) are busy to the point where pharmacists do not have enough extra staff and time to stop and research a question that quickly. Yet, if the patient situation is truly urgent, then measures should be taken by the staff to perform a quick and fairly complete search through tertiary, secondary, and primary literatures with follow-up and additional searching to be performed after the initial information is conveyed to meet the given time frame.

In terms of appropriate health literacy, one would not respond to a patient asking about a possible drug-drug interaction in the same way as one would respond to a physician. It is commonly understood that patients and the general public read at an 8th to 10th-grade level. However, surveys and guidelines from governmental bodies (e.g., Health Literacy Innovations' National Survey of US Medicaid Guidelines,[11] CDC Health Literacy Tools[12]) suggest that medical writings should be geared toward a sixth-grade reading level. In practice, it is very difficult to write patient information at a sixth-grade reading level; however, using shorter sentences, smaller words, plain language, and not using medical jargon can help patients better process and understand the information they receive. Software for editing and creating documents (e.g., Microsoft Word) usually has a tool to show readability statistics (e.g., Flesch-Kincaid Grade Level) on review.

Retrieving Necessary Background Information

Most drug information requests have to do with patient-specific situations. For example, a patient may be experiencing an adverse drug reaction (ADR) to an angiotensin-converting enzyme inhibitor. To help the patient, information needs to be gathered concerning the nature and severity of the reaction, previous drug allergies, and when the medication was taken in relation to the appearance of the ADR. For completeness in providing an accurate and helpful response, other basic information should be collected (Table 11.3).[7,8,10] If working in a hospital or clinic, it may be easy to obtain such information with a medical record number.

To remember the various information needed or questions that should be asked, it may be helpful to develop a call sheet (Fig. 11.1). This can be placed near the phones in a pharmacy or call center and can be easily filled out and completed by staff, residents, and students when receiving requests to provide drug or medical information. Remember, if a valuable piece of information is missed during the initial communication, it is usually appropriate to contact the requester again and ask for further information.

Determining the True Informational Need and Categorizing the Ultimate Question

Once the necessary background information has been collected, the final thing to do before actually starting to research the question would be to come up with the requester's true informational need.[8,10] In other words, what question are they really asking or what do they really want to know? This is extremely important because the "worst" thing about having to research a lengthy and in-depth drug information question is to have to do it again. Therefore, it is a good idea to repeat back to or clarify with the requester the interpretation of the question being asked.

Next, it is necessary to classify the question into one of the following categories (Table 11.4). Categorizing the question will determine a focus for research, as well as offer a concentrated set of references to help answer the question. For example, if a nurse asked if heparin, phenylephrine, and milrinone could be run together via a triple-lumen catheter, categorizing the question to "IV compatibility" would lead to initially using tertiary references such as the *Handbook on Injectable Drugs* (i.e., Trissel's) or the *King Guide to Parenteral Admixtures*. That would also eliminate the need to search

TABLE 11.3 Important Background Information to Be Collected

Age, sex, height/weight	Medication history • All over-the-counter and alternative products/supplements • Dose/duration of therapy
Drug/food allergies • Nature of allergy/reaction	Current disease states and diagnoses
Past medical history	Pertinent/current laboratory values
Major organ function (e.g., liver, kidneys)	Diluents/rates of administration (if intravenous)

Drug Information Call Sheet

Caller Name: PharmD MD Nurse Other

Affiliation:

Location:

Phone number:

Email:

When do you need this information:

Can you tell me more about the situation:

Is there anything special that I should know about this patient:

Question:

Other info (Labs, Medication, Allergies, Medical history):

FIGURE 11.1 Example of a drug information call sheet.

TABLE 11.4 Common Categories for Drug Information Questions

Adverse Drug Reaction	General Information
Compounding	Method of administration
Compatibility of IV medications	Patient education
Contraindications/warnings	Pharmacokinetics
Dietary supplements, herbal and homeopathic medications	Pharmacology
Dosage/schedule	Pediatrics
Dosage form/formulation	Pregnancy/lactation
Drug allergy	Pharmacoeconomics
Drug availability/cost	Reference/citation search
Drug–drug/drug–food/drug–lab/drug–disease interactions	Stability/storage
Drug identification	Therapeutic consult/indications
Formulary management	Toxicology

for references that would not necessarily be helpful or specific enough (e.g., *Facts and Comparisons monographs*, *Goodman and Gilman's: The Pharmacological Basis of Therapeutics*).

At this point, the demographics of the requester have been identified, appropriate background information has been collected, the true informational need has been assessed, and the question has been correctly categorized. Now, it is time to begin the initial stages of answering the question. It all starts with building an appropriate searching strategy.

Developing a Sound Searching Strategy and Conducting a Thorough Search (Validate and Verify)

Tertiary Resources

Once a clinical question is formed, it is time to search for the best evidence. The easiest place to start a search is in tertiary resources. Tertiary resources provide a broad overview of a topic and present information in an easy-to-read format, often with supporting references that lead to primary articles. Examples of tertiary resources include UpToDate, LexiComp, Micromedex, and Facts and Comparisons. Most online tertiary resources will indicate the date when a literature review was last performed on the summary topic and the date when the summary was last updated. These two dates may differ and may not always result in the topic summary being updated (i.e., if no new practice-changing information had been published). If in print, one must look at the publication date and determine how relevant the information might be at the time of research. The most reliable tertiary resources are evidence-based rather than based on expert opinion. Narrative reviews are considered a tertiary resource, whereas systematic reviews are a primary resource (e.g., new statistics/data are generated). Evidence-based summaries will include levels of evidence throughout, and the resource should clearly describe how it determines the quality and levels of evidence.

Secondary Resources

If one cannot find an appropriate or detailed-enough answer to a clinical question in tertiary resources, then secondary resources should be considered. Secondary resources—such as Cochrane Database of Systematic Reviews, PubMed, Scopus, Embase, and Web of Science—compile primary literature such as journal articles, technical reports, and conference papers.

When searching for secondary resources, it is essential to develop a comprehensive search strategy to retrieve all relevant articles that answer the clinical question. There are several techniques that will ensure that a search strategy is well formulated.

First, use subject headings, when available. For example, PubMed includes the MeSH Database. MeSH (Medical Subject Headings) is the National Library of Medicine's controlled vocabulary thesaurus used to index articles in PubMed. Many other databases include subject-heading thesauri, as well. Searching with subject headings ensures that everything indexed on that topic will be retrieved, regardless of what keywords are used in the articles. For example, searching with the PubMed MeSH heading, "Urinary Tract Infections," will retrieve articles that use synonyms (e.g., UTI) as well as narrower terms (e.g., bacteriuria, pyuria, schistosomiasis hematobia) for urinary tract infection. Depending on journal impact factors and other considerations, it may take several months for an article to be assigned subject headings. Therefore, one drawback to searching exclusively with subject headings is that the newest articles may not be retrieved through the search. Using Boolean operators to combine subject headings with keywords will result in a comprehensive search strategy that retrieves both relevant and recently published articles.

Boolean operators (e.g., AND, OR, NOT) are used to connect and define relationships between your search terms (Fig. 11.2).[13]

To frame a search, first list the main concepts within the PICO format (i.e., patient/population/problem, intervention, comparison, outcome). Combine search concepts with "AND" to narrow the number of articles retrieved. List synonyms for each concept and combine synonyms with "OR." Combining synonyms with "OR" will broaden the search and retrieve more results. For the clinical question, "In women with uncomplicated urinary tract infection, what is the effect of nitrofurantoin on clinical cure rate?" a search strategy may look like ("Urinary Tract Infections"[MeSH] OR urinary tract infection) AND ("Nitrofurantoin"[MeSH] OR Macrobid).

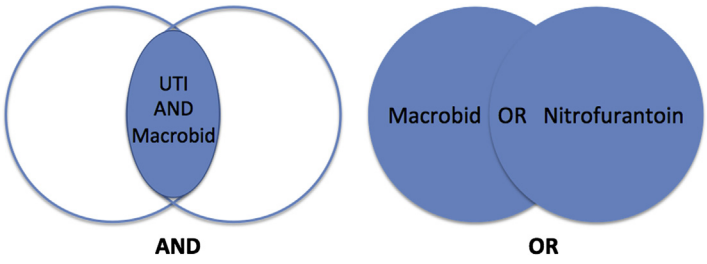


FIGURE 11.2 Boolean operators.[13]

To further focus the search, use subheadings in conjunction with subject headings. Subheadings are used to describe the specific aspects of the subject heading that are pertinent to the article. Not all databases that employ subject headings will use subheadings, but of those that do, most subject headings will be indexed with one or more subheading. A subject heading using a subheading, for example, "Urinary Tract Infections/drug therapy"[MeSH], will be more focused and retrieve fewer, but more relevant, results.

Another search technique is to use quotes and truncation. Put quotes around words to search them as an exact phrase. Put an asterisk at the root of a word to search alternate endings. For example, hospital* would retrieve the words hospital, hospitals, and hospitalized.

The use of subject headings, Boolean operators, quotes, and truncation, will ensure that the search strategy is well formed. After reviewing the results of the search, filters—or limits—can further refine search results. Filters differ among databases. Common filters may include publication dates, article types (e.g., systematic reviews, clinical trials), age, species (e.g., human, other animals), language, and many more. Filters can quickly reduce the search results to a reasonable number of relevant citations. Because searching is complex and vital to EBP, it is imperative to learn the best strategies and consult with a librarian when necessary, to ensure that all relevant articles are being retrieved.

Primary Literature

One must perform a comprehensive search of secondary resources to identify and retrieve primary literature. Primary literature consists of original research, either published or unpublished, such as systematic reviews, RCTs, or case reports. Not all primary literature is created equally. The Hierarchy of Evidence Pyramid illustrates the strength of study design (Fig. 11.3).

Systematic reviews and metaanalyses have the least biased study design and strongest strength of evidence. One systematic review or metaanalysis may synthesize the findings of many RCTs and postulate new conclusions about specific therapies. Therefore, it is easier for the information seeker to read one systematic review rather than appraise the findings of many RCTs. Narrative reviews differ from systematic reviews and metaanalyses in that because they are not rigorously designed; there is more likelihood of bias (Table 11.5).

RCTs, cohort studies, case-controlled studies, cross-sectional studies, case reports, and case series differ in their strength of study design. Generally, it is best to aim for the study design that is highest on the pyramid, but appropriate study design depends on the question. Both the Hierarchy of Evidence Pyramid and Best Study Design chart are meant to be guides to choosing the best study design for a particular question being answered (Table 11.6).

To obtain the primary literature, one must gain access through an institutional license unless the literature is freely available to the public (e.g., open access, public access). Open access articles are free to use without an institutional subscription and are often free of most copyright and licensing restrictions. Public access articles are also free to use without an institutional license and consist of research funded by the United States' National Institutes of Health and other government agencies. PubMed Central is a free database that archives publicly accessible full-text science articles. Those who are affiliated with an institution can gain access to library-subscribed full-text articles or can request articles through their library's interlibrary loan service. The entire search process—from forming the clinical question to developing the search strategy to obtaining full-text articles—is essential to answer the clinical question at hand properly.

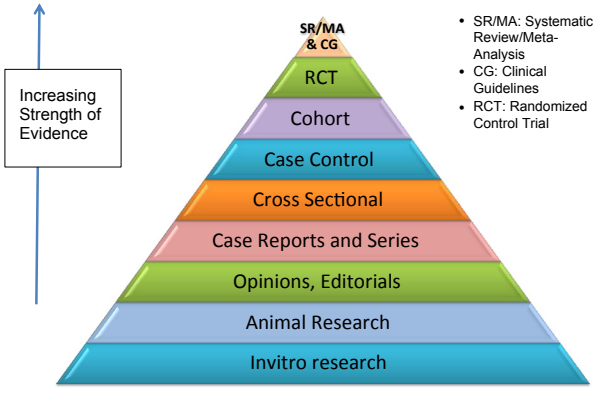


FIGURE 11.3 Hierarchy of evidence pyramid.[14]

TABLE 11.5 Comparison of Types of Review Articles

	Description	Advantages	Disadvantages
Narrative review	Authors' opinions (usually a tertiary reference)	Gain expert's insight, readable, broad scope	Potential bias toward author's opinion, reject alternative viewpoints, rarely reveals its sources of information, may not include the most up-to-date references if the author was not aware of them
Systematic review	Standardized retrieval, appraisal, and summary of the literature (could be tertiary or primary, depending on if new statistics were run and new information was presented)	Narrow clinical focus, best evidence, nonbiased, comprehensive literature critique	Available literature (i.e., publication bias), criterion selection may be too strict (limits generalizability) or too broad (hard to compare studies), search strategy may not be comprehensive
Metaanalysis	Systematic review with the statistical synthesis of data (usually primary b/c new statistics are performed to analyze data and come up with new/alternative recommendations)	Increases the power of the analyses, influence practice earlier	Poor translation to individuals, overinflated findings, limited to previous publications (i.e., publication bias), statistical results may lose meaning if studies that are combined are highly variable

TABLE 11.6 Best Study Design for Question Type

Question Type	Study Design (first being the best)
Diagnosis	Prospective, blind comparison to a gold standard
Therapy	Randomized Controlled Trial → Case Control → Case Series
Prognosis	Cohort Study → Case Control → Case Series
Harm/etiology	Randomized Controlled Trial → Cohort Study → Case Control → Case Series
Prevention	Randomized Controlled Trial → Cohort Study → Case Control → Case Series

Evaluation, Analysis, and Synthesis of Gathered Information

It is important to summarize the research findings in a format that promotes ease of use for the requester. The summary should quickly and straightforwardly provide *actionable steps* to be taken based on the results of the literature search, with supporting evidence for the recommended action. To ensure that the information is always displayed in a concise, consistent format, it may be helpful to develop an evidence summary template. Important items to include in this template would be as follows:

- Clinical question of interest using PICO format
- Therapeutic recommendation
- Evidence-based rationale
- References

When a requestor contacts a pharmacist or drug information center, their goal is to understand "What should I do?" in a specific clinical situation. It is helpful to keep this in mind when responding to their request. It is important to begin responses by formalizing the clinical question and providing an initial therapeutic recommendation. This should be followed by background information from either tertiary and primary references that support the conclusion/recommendation. Finally, some basic monitoring parameters or side effects for the clinician or patient to be aware of and a short summary of the major highlights of the information should be provided at the end of the response. This allows the requestor to quickly know "what to do," as well as being able to view the response in a comprehensive format that can easily be saved for later reference or placed in the patients' chart.

The depth of the evidence-based rationale for a recommended therapy will hinge on two main factors: urgency of the response and patient-specific factors. For example, in urgent situations where timely response is critical, focusing on

tertiary data sources and evidence-based guideline recommendations to supply the rationale for decision-making saves time without affecting clinical integrity. However, when making therapeutic recommendations in complex patients, it becomes important to rely on secondary resources and primary literature (i.e., systematic reviews, RCTs) to ensure more precise decision-making. In most cases, though, it is important to balance these two factors against each other when deciding what sources to include in the response.

The content of the evidence-based rationale should highlight the various sources available for the clinical question:

- Tertiary resources and evidence-based guidelines
  The tertiary resources and narrative review articles can be useful sources of evidentiary support when time is a limiting factor for searching the primary literature. Because these sources already provide a broad overview of a clinical topic, the summary is largely completed. Where appropriate, highlight the pertinent evidence cited in these resources and focus on tailoring the summary to the requestor and their potential health literacy needs.
  It is also appropriate to use evidence-based guidelines to formulate a therapeutic recommendation if they exist. However, it is important to review a guideline first to determine how much trust can be placed in its recommendations. Many guideline issuers assign a "strength" or quality of evidence to support their decision-making, so understanding how that assignment was made is critical. Additionally, it is important to look at the development team and any potential conflicts of interest that may exist. The specific criteria for determining the trustworthiness of a guideline are discussed later in this chapter.
- Primary literature
  Secondary resources (i.e., PubMed, Scopus) contain primary literature in the form systematic reviews, metaanalyses, or original research studies (i.e., RCT, observational studies, case studies). The strongest study design for a therapeutic intervention is an RCT, but that does not mean that observational studies cannot assist with decision-making. Regardless of the source, it is essential to critically analyze the publication with three key questions in mind:
  - *Are the results valid?* It is important to critically analyze primary research to identify any significant design limitations, as these limitations impact the level of confidence that can be placed in the results (Table 11.7). For example, when reviewing a RCT regarding the efficacy of lorazepam (Ativan) compared with diazepam (Diastat) as a first-line therapy for pediatric patients in status epilepticus, it is important to evaluate factors such as sample size, blinding, allocation concealment, and loss to follow-up when determining the validity of the study's results.
  - *Are the results significant?* Use the statistical findings of the study to evaluate the significance or importance of the results, with a focus on the size and precision of the treatment effect (e.g., an estimate of effect, 95% confidence intervals, *P*-values). Common statistics presented as an estimate of effect, used in therapeutic intervention studies

TABLE 11.7 Assessing the Validity of Research Studies

Study Type	Questions for Evaluating the Validity
Systematic reviews and metaanalyses	• Did the review address a focused clinical question? • Was the search detailed and exhaustive? • Were the studies of high methodological quality? • Was study inclusion evaluated by multiple authors?
Randomized controlled trials and quasiexperimental studies	• Was the sample size large enough to reflect statistical significance (i.e., power calculation)? • Were the subjects randomized, and if so was it concealed? • Were the subjects, providers, and outcome assessors blinded? • Were the groups similar in terms of demographic and prognostic factors? • Was intent-to-treat analysis of outcomes used? • Was follow-up complete and sufficient in length to assess for study outcomes? • Were results reported for all primary and secondary outcomes?
Observational studies	• Was the sample size large enough to reflect statistical significance (i.e., power calculation)? • Was the data collected prospectively? • Were study participants representative of a defined population (with or without controls)? • Were all important confounding factors identified and measured? • Was follow-up complete and sufficient in length to assess for study outcomes?

include the following: relative risk reduction, absolute risk reduction, and number needed to treat/harm. The precision of these findings is based on a range of values presented as a 95% confidence interval. For example, a narrow confidence interval range represents a more precise result.

- *Will these results help determine care for the patient?* A study's results can be generalized to a specific patient most of the time, but it might also be useful to assess whether or not the patient in question would qualify for the study based on its inclusion and exclusion criteria. In addition, how important are the results in terms of clinical significance? For example, if a study only shows a 0.5% improvement in blood glucose (hemoglobin A1C) over 6 months, do those results seem important clinically (i.e., is the benefit big enough)?

Formulate and Communicate the Response[8,10]

One thing to consider as the response is being written is how the requester will use the information. Because most drug information questions are related to one patient, the information provided will most likely be used for that specific scenario without further application. However, if a written response provided to a clinician is included in the patient's chart, then that may become part of the official medical record. It would be important to determine the end result of the information before sending the response to the requester; however, if one has completely followed the systematic approach with the inclusion and proper documentation of references/databases searched, all aspects of research should be covered and easily reproduced if there is ever a question (e.g., clinical, legal) regarding the information provided.

Conversely, drug information centers associated with hospitals tend to receive requests from various departments to create tables and charts that will be used by other healthcare professionals in taking care of a group of patients. For example, the creation of an insulin comparison chart might be used by a diabetic management service as a reference guide to the differences between insulin products (e.g., onset of action, duration of action, product availability). Therefore, for requests that will affect a larger group of patients, extra care must be taken to validate and verify all information and document all references where specific information is located.

Something else to consider would be the format in which the requester would prefer the information (e.g., table, chart, written text). Additionally, should the answer be sent via email or would the requester rather receive a call with a discussion of the information provided? Some of that may be up to the judgment of the person doing the research. For example, if one is researching a question that can best be answered with primary literature and four pertinent studies have been found, it might be best to summarize those in a chart and email the document to the requester. If they have any questions about the summary, the offer should be made to follow up at a later time.

Conduct Follow-Up/Provide Documentation[8,10]

Follow-up for most questions is not always necessary; however, it can be done as a courtesy. Most clinicians will welcome a call 2 or 3 days after the information was given to determine if the information was helpful or not. Follow-up could also be used as self-survey or process improvement tool to assess the overall service and information provided.

As mentioned earlier, appropriate documentation (e.g., citations, databases searched, search terms used) should be recorded in whatever database or archive system is being used to store all of the drug information questions answered by the clinician or service. In addition, if a written response is being provided, the complete answer along with the references and databases used to create the response should be included and provided to the requester. Appropriate documentation of references within the text will come from various medical writing styles (e.g., American Medical Association, Chicago, American Psychological Association); however, the correct way to reference a body of work (e.g., article, book, Internet database) is often standardized. That information can be found within the "how to submit an article" section of most journals, as well as the National Library of Medicine's *Citing Medicine.*[15]

OBJECTIVE 11.3. DESCRIBE THE PROCESS OF PREPARING SYSTEMATIC REVIEWS USING EVIDENCE-BASED PRACTICE

In addition to evidence-based individual decision-making, EBP methods can also be used to answer a specific clinical question for a group of patients (i.e., systematic reviews). Systematic reviews seek to answer a focused clinical question with a comprehensive literature search, strict inclusion and exclusion criteria, and team-based appraisal of evidence. "Researchers conducting systematic reviews use explicit methods aimed at minimizing bias, in order to produce more reliable findings that can be used to inform decision-making."[16] The systematic review process follows the five steps of

EBM (i.e., ask, search, critical appraisal, implement, and evaluate) but with more rigor, thus leading to less likelihood of bias. However, like primary literature, not all systematic reviews are created equally.

The most common types of review articles are systematic reviews, metaanalyses, and narrative reviews. Many other less common review types often get confused with systematic reviews.[17] Often clinicians will decide that they want to write a systematic review until they realize the time and effort that goes into developing one. According to the Cochrane Collaboration, the timeline for a systematic review is 12 months.[18] The Cochrane Collaboration is an international organization that is a leader in developing high-quality, evidence-based health sciences systematic reviews. The Cochrane Library publishes these systematic reviews in the Cochrane Database of Systematic Reviews, which also includes other types of reviews and publications.

The PRISMA (Preferred Reporting Items of Systematic reviews and Meta-Analyses) Statement, which consists of a checklist and flow diagram, was published to standardize the development of systematic reviews.[19] The 27 PRISMA checklist items guide authors through the systematic review process and is recommended for all authors embarking on the review process.[19] Before committing to a systematic review, several things must be accomplished. First, the team of experts who will work on the systematic review should be identified. A team-based approach is essential to establish inter-rater reliability and reduce bias. That team should identify the population/problem, intervention, comparison, outcome, and how benefits and harms will be measured. The specific question being researched should occur in the context of existing knowledge and literature. Therefore, when writing a systematic review, the authors should provide background on the current body of literature and explain what new knowledge their review provides.

The authors should then determine specific inclusion and exclusion criteria. These criteria may relate to details about the study population such as age range, setting, or point in the disease or may relate to study design, published versus unpublished research, and other limitations. The team may consider registering their systematic review in PROSPERO,[20] an international prospective register of systematic reviews, or with the Cochrane Review Group[21] if they are writing a Cochrane Systematic Review. The protocol is published as a placeholder until the full review is completed. The protocol can be compared to the final systematic review to determine if any bias or discrepancies have occurred.[22]

Arguably the most important step in the systematic review process is searching the literature. Authors should collaborate with a librarian to ensure the most comprehensive search possible. Librarians will search a variety of resources, and search strategies should be highly sensitive so that all relevant articles are retrieved. Librarians will combine subject headings and keywords to retrieve everything on the topic including narrower topics and articles electronically published ahead of print. Other relevant articles may be discovered through hand-searching reference lists, reviewing citing articles, and personal communication with key authors. The librarian will keep track of the reproducible search strategy, dates searched, number of results, filters applied, and other details for each database. This comprehensive search strategy is the core of the systematic review and will be described in detail in the methods section of the paper and also published as an appendix.

Once the literature review is complete, references are normally exported into a reference manager such as EndNote[23] or RefWorks.[24] The reference manager will be used to eliminate duplicate citations and to properly cite and format references in the final manuscript. The references will then be screened by at least two team members for inclusion or exclusion. There are many review managers available to facilitate the screening process, such as RevMan[25] or Abstrackr,[26] but Microsoft Excel[27] can be used, as well. The titles and abstracts should be screened first to remove references that are obviously irrelevant. The next step is to retrieve the full text of any potentially relevant articles (the librarian can help with this). At least two authors will independently read the full-text articles to determine if they meet the inclusion and exclusion criteria and vote on whether or not the article should be included in the systematic review. They will then discuss conflicting votes and decide on the final evidence to be included. The PRISMA Flow Diagram, which illustrates the number of records identified, screened, included, and excluded, should be included in the systemic review (Fig. 11.4).[19]

The next step in the systematic review process is to summarize the evidence and interpret the findings. The PRISMA Statement provides a checklist of items to consider when reporting results including (1) study selection, (2) study characteristics, (3) risk of bias within studies, (4) results of individual studies, (5) synthesis of results, (6) risk of bias across studies, and (7) additional analysis.[19] There are many tools available, such as the Cochrane Risk of Bias Tool, to assess considerations such as the risk of bias (i.e., validity or methodological quality of studies).[28] The characteristics of included studies should be detailed in a table and if study methods are too heterogeneous, authors should clearly explain why a metaanalysis cannot be done. Authors should, at a minimum, qualitatively evaluate the included studies and identify what future research is needed to fill existing gaps in knowledge or strengthen the body of evidence. Systematic reviews should be updated regularly or include commentary to explain why this is not possible.[16]

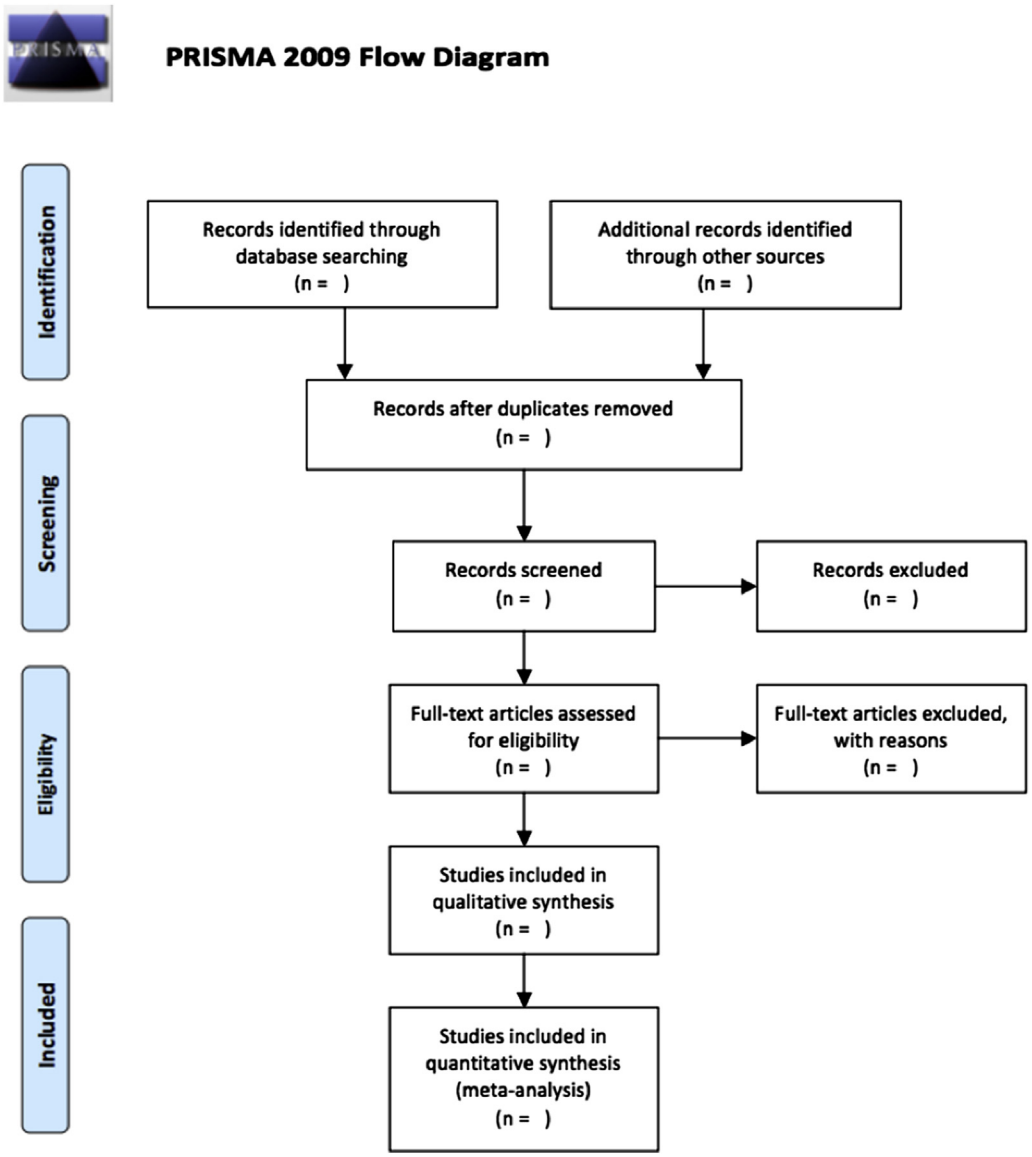


FIGURE 11.4 PRISMA flow diagram.[19]

CONCLUSION

Overall, when responding to questions from clinicians about patient-specific instances, effective and efficient research, analysis, and use of EBP/medicine are all extremely important tools to have. Use of the modified systematic approach to answering drug information and medical-related questions is a commonly employed process that yields positive results. Development of a proper searching strategy will yield quality resources and data. Results should be validated and verified in more than one source, as various references may be out of date or contain different information. Correct analysis and synthesis of gathered material is also necessary, along with finding and reviewing strong EBP (e.g., RCTs, systematic reviews) when offering recommendations to practitioners. Finally, clear communication of the answer, as well as supporting information, is essential for the pharmacist to provide to the end user for ultimate application to the patient.